# User Manual

BigBrotherProMax

## 1 Overview

BigBrotherProMax is a BLE-based sleep monitoring system. An ESP32 pulse oximeter and a LilyGo T-Wrist watch stream heart rate, $SpO_2$, and accelerometer data over Bluetooth to a Raspberry Pi. The Pi captures both data outputs, runs sleep analysis, and displays results through a local web dashboard.

Source code: `https://github.com/Sevi-S/BigBrotherProMax/`

## 2 Hardware

- Raspberry Pi (with Bluetooth)
- LilyGo T-Wrist watch
- Embedded SPO2 and heart rate sensor
- ESP32 pulse oximeter module

## 3 Flashing the Firmware

Both the ESP32 oximeter and the LilyGo watch run custom C++ firmware that must be compiled and uploaded before first use.

1. Connect the LilyGo watch and the ESP32 module to your computer using Micro-USB.
2. Make sure the embedded sensor is connected to the ESP32 prototype correctly.
3. Open Arduino IDE.
4. Copy or open the project code from the GitHub repository.
5. Install the required libraries.
6. Select the correct board and port for each device.
7. Set the required flash settings in Arduino IDE (see Figure **??**).
8. Compile and upload the firmware to the ESP32 oximeter.
9. Compile and upload the firmware to the LilyGo watch.
10. Verify that the watch UI appears correctly on the screen.

Once flashed, both devices advertise over BLE automatically on power-on.

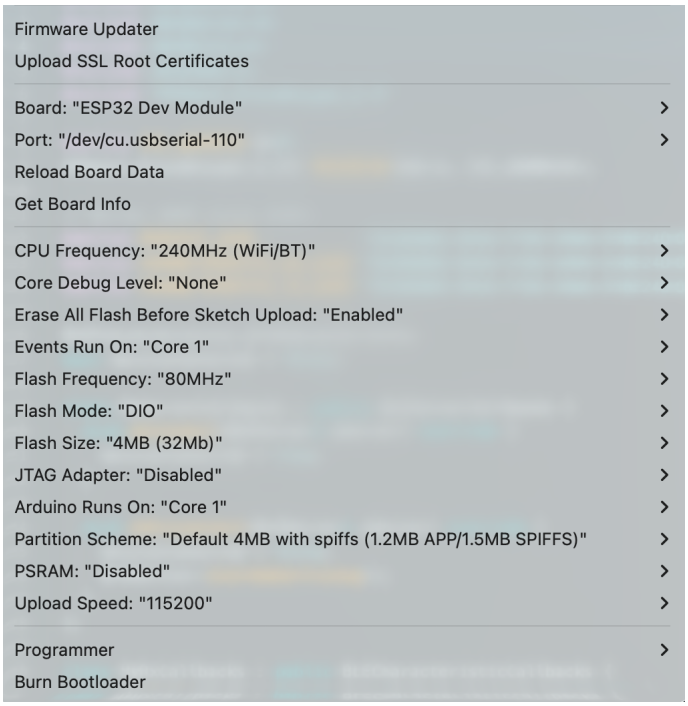


Figure 1: Arduino Flash Settings

# 4 Raspberry Pi Setup

```
git clone <repo-url>
cd BigBrotherProMax
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Ensure Bluetooth is enabled on the Pi (check via the system tray or `bluetoothctl`).

# 5 Running

```
source .venv/bin/activate
./start.sh
```

This launches the BLE capture script, the Flask web UI, and opens Firefox to `http://127.0.0.1:8000`.

Press **Ctrl+C** to stop all processes.

# 6 Usage

1. Power on both the oximeter and the watch.
2. After rebooting the Raspberry Pi, go to the project directory and activate the Python virtual environment using `source .venv/bin/activate`.
3. Run `./start.sh` on the Raspberry Pi.
4. The script scans for both BLE devices and connects automatically.
5. Press the touchscreen **Start Tracking** button on the watch to begin recording.
6. A session begins immediately after the button is pressed and data starts streaming.

7. During recording, the watch continuously sends accelerometer data and the oximeter sends HR and $SpO_2$.
8. After 30 seconds of no watch data, the session ends automatically and analysis starts.
9. Open the dashboard to view the sleep score, HR/$SpO_2$ charts, and sleep stages.

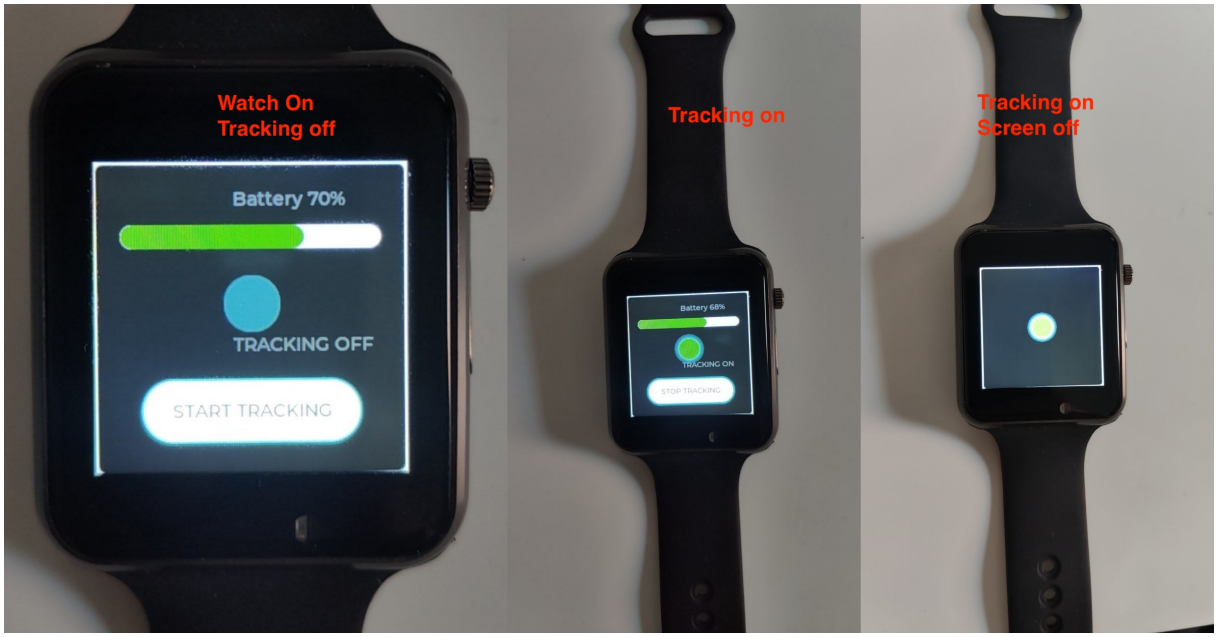


Figure 2: Watch face

## 7 Utilities

`python3 cleardb.py` Delete today's session from the database.
`python3 cleardb.py --full` Delete and recreate the entire database.

## 8 Troubleshooting

- Devices not found: Make sure Bluetooth is on and both devices are powered. The script retries every 5 seconds.
- No HR/$SpO_2$ data: Ensure the oximeter sensor is in contact with skin. Values of $-1$ indicate no reading.
- Database errors: Run `python3 cleardb.py --full` to reset.
- UI won't open or unpredictable behavior: Run `ps aux | grep python` to check for stale processes that didn't exit, and force kill them with `kill -9 <pid>`.